

Modeling nonlinear relationships

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Polynomial models

Log transformations

Log-log models and price elasticity



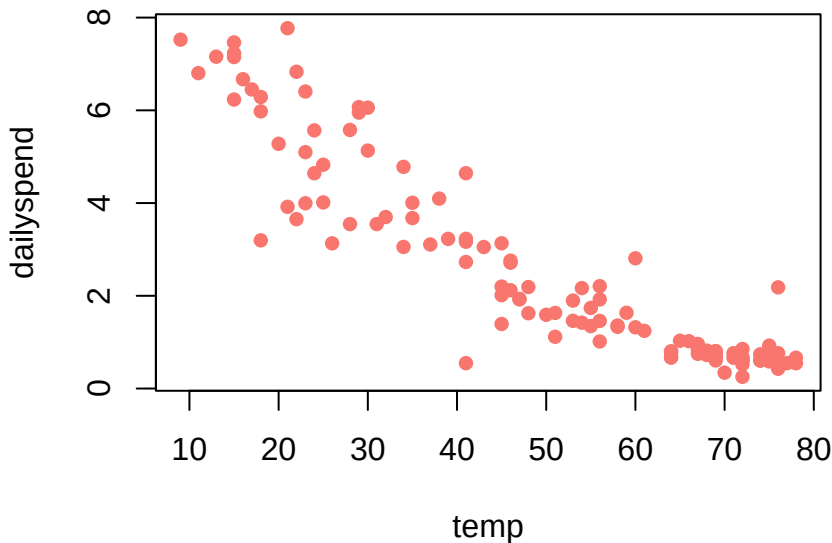
Utilities data set

The data set `utilities` contains information on the utility bills for a house in Minnesota. We'll focus on two variables:

- **`dailyspend`** is the average amount of money spent on utilities (e.g. heating) for each day during the month
- **`temp`** is the average temperature outside for that month



What problems do you see here?



```
model <- lm(dailyspend ~ temp, data=utilities)
summary(model)
```

Call:

```
lm(formula = dailyspend ~ temp, data = utilities)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.847	-0.504	-0.024	0.515	2.448

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.34762	0.20645	35.6	<2e-16 ***
temp	-0.09643	0.00391	-24.7	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

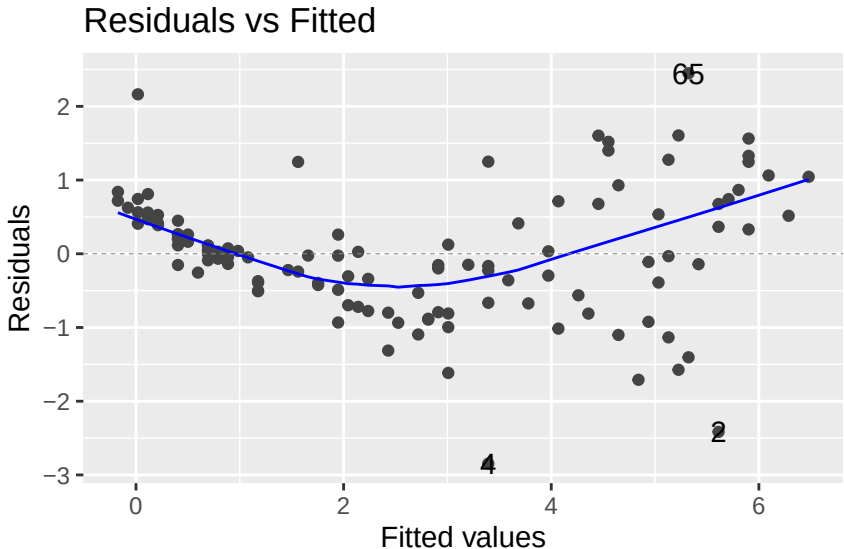
Residual standard error: 0.866 on 115 degrees of freedom

Multiple R-squared: 0.841, Adjusted R-squared: 0.84

F-statistic: 608 on 1 and 115 DF, p-value: <2e-16



Linearity and homoscedasticity are violated





Using polynomial regression to fix problems

- If a polynomial curve (e.g., quadratic, cubic, etc) would be a better fit for the data than a line, we can fit a curve to the data
- The way we do this is by adding X^2 , X^3 , etc. terms to the model as additional predictors

```
poly.model <- lm(dailyspend ~ temp + I(temp^2), data=utilities)
summary(poly.model)
```

Call:

```
lm(formula = dailyspend ~ temp + I(temp^2), data = utilities)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.8725	-0.2805	-0.0393	0.2639	2.1912

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.472289	0.390789	24.24	< 2e-16 ***
temp	-0.211555	0.019105	-11.07	< 2e-16 ***
I(temp^2)	0.001248	0.000204	6.12	1.3e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

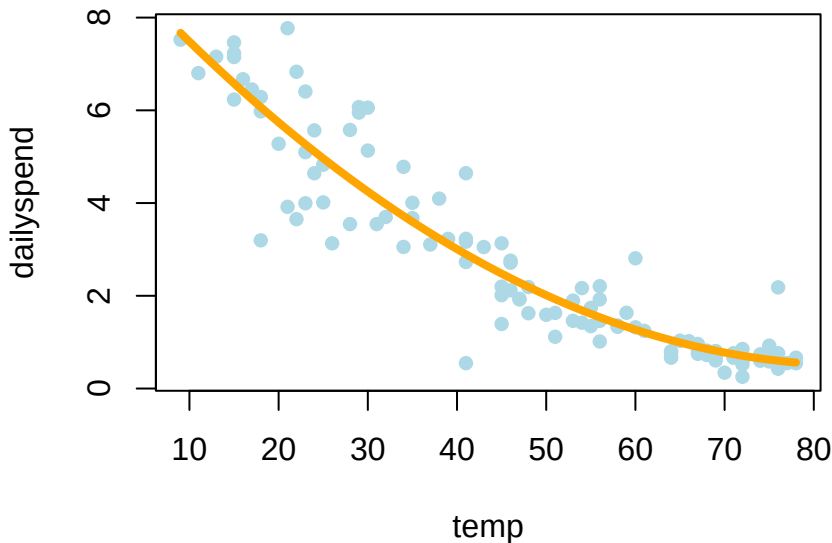
Residual standard error: 0.755 on 114 degrees of freedom

Multiple R-squared: 0.88, Adjusted R-squared: 0.878

F-statistic: 419 on 2 and 114 DF, p-value: <2e-16

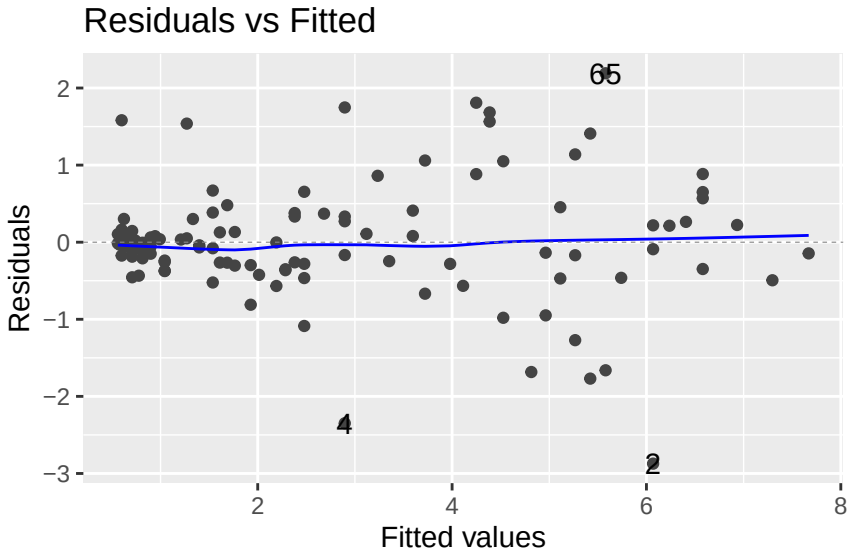


Adding an X^2 term fits a parabola to the data





It solves the linearity problem!





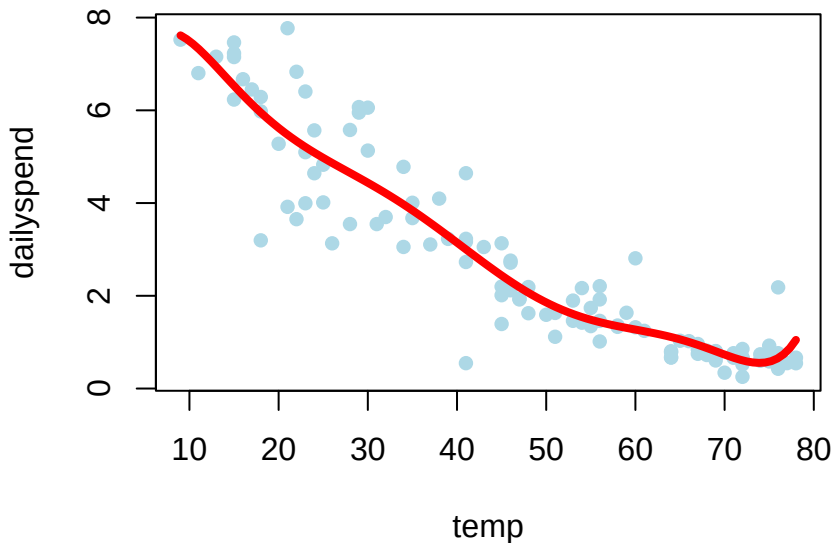
What about a higher-order polynomial?

- We could fit a cubic curve by adding an X^3 term (in addition to the X and X^2 terms)
- Making the polynomial higher order will increase R^2 —why not go nuts and fit a 7th degree polynomial?

Degree		R^2
1	linear	0.841
2	quadratic	0.8803
3	cubic	0.8812
4	quartic	0.8813
5		0.8815
6		0.8822
7		0.8837

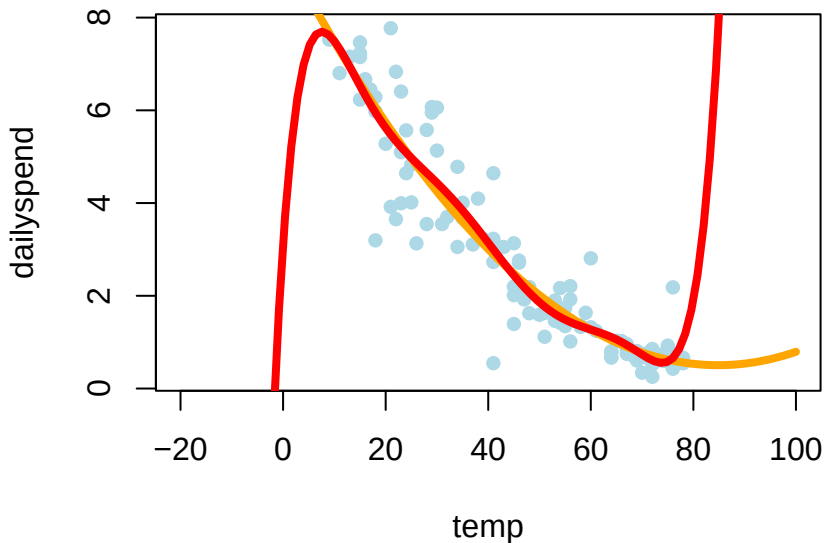


Too high a degree “overfits” the data





Too high a degree creates dangers with extrapolation





- Start simple: only add higher-degree terms to the extent it gives you a substantial increase in R^2 , or makes an assumption hold that wasn't holding before



Building polynomial models

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- Be careful about overfitting



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- Be careful about overfitting
- Be particularly careful about extrapolating beyond the range of the data
- Mind-bender: We can think about an X^2 term as an interaction of X with itself: in a parabola, the slope depends on the value of X !

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Using transformations to fix problems

- Sometimes, a violation of regression assumptions can be fixed by transforming one or the other of the variables (or both).
- When we transform a variable, we have to also transform our interpretation of the model
- Often the new interpretation is more meaningful!

LOG!





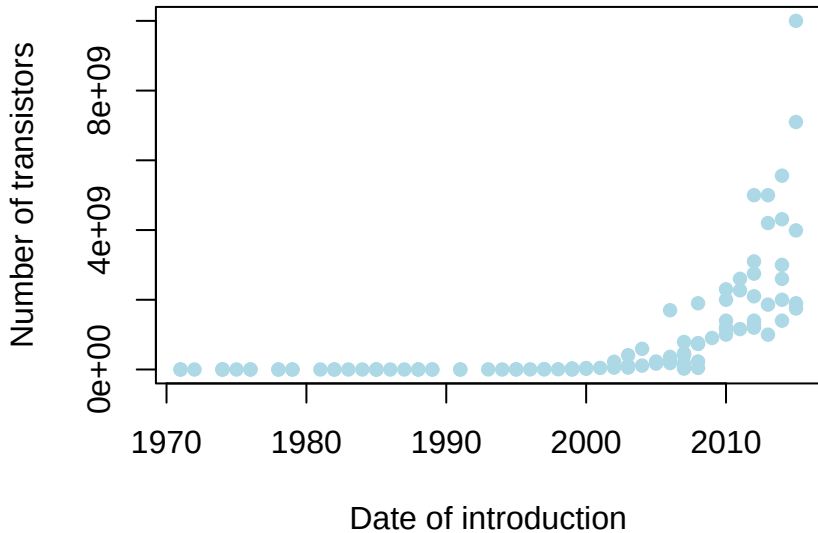
The log transformation

The **log** transformation is frequently useful in regression, because many nonlinear relationships are naturally exponential.

- $\log_b x = y$ when $b^y = x$
- For example, $\log_{10} 1000 = 3$, $\log_{10} 100 = 2$, and $\log_{10} 10 = 1$
- The natural log is $\log_e$, where $e \approx 2.72$ — when we say “log” we will usually mean “natural log” (although for our purposes the base doesn’t matter)

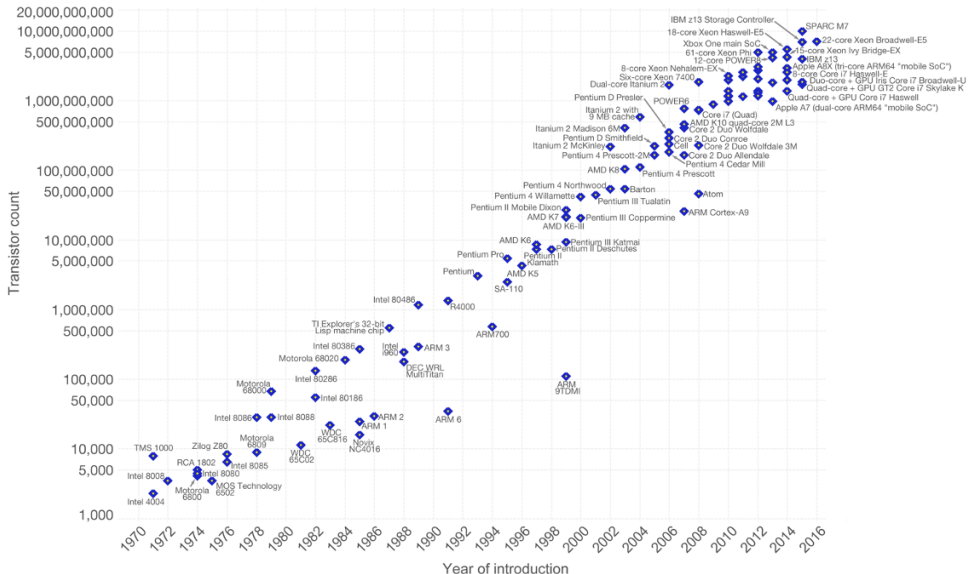


- Moore's Law was a prediction made by Gordon Moore in 1965 (!) that the number of transistors on computer chips would double every 2 years
- This implies **exponential** growth, so a linear model won't fit well (and neither will any polynomial)



Moore's Law – The number of transistors on integrated circuit chips (1971-2016)

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are strongly linked to Moore's law.



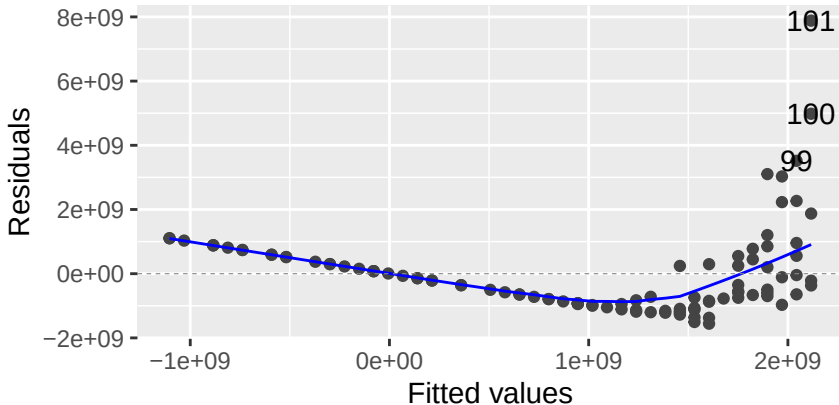
Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)
The data visualization is available at [OurWorldinData.org](https://www.ourworldindata.org). There you find more visualizations and research on this topic.



A linear model is a spectacular fail

```
linear.model <- lm(Transistor.count ~ Date.of.introduction,  
                  data=moores.law)  
autoplot(linear.model, which=1, ncol=1)
```

Residuals vs Fitted

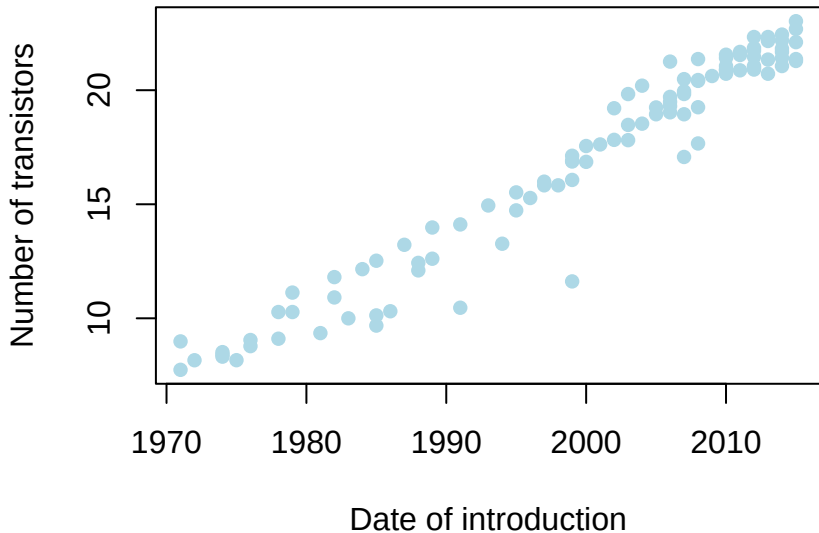




- If $Y = ae^{bX}$, then $\log(Y) = \underbrace{\log(a)}_{\beta_0} + \underbrace{b}_{\beta_1} X$
- In other words, $\log(Y)$ is a linear function of X
- So to model Y as an exponential function of X , predict $\log(Y)$ as a linear function of X



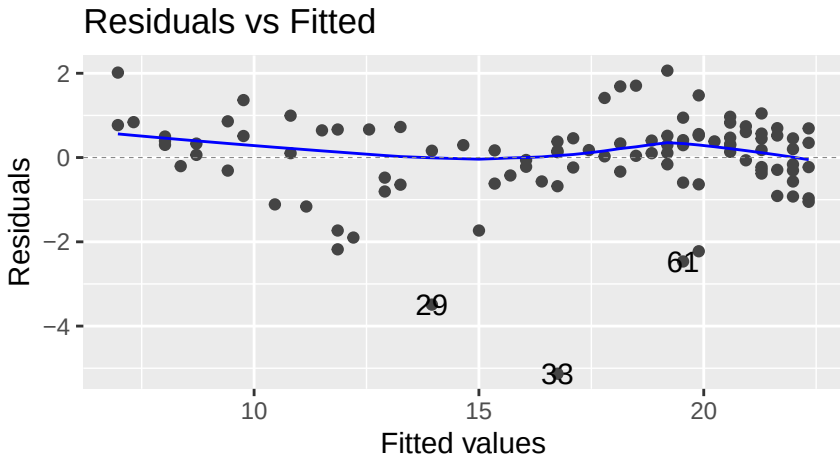
$\log(\text{Transistors})$ does have a linear relationship with year





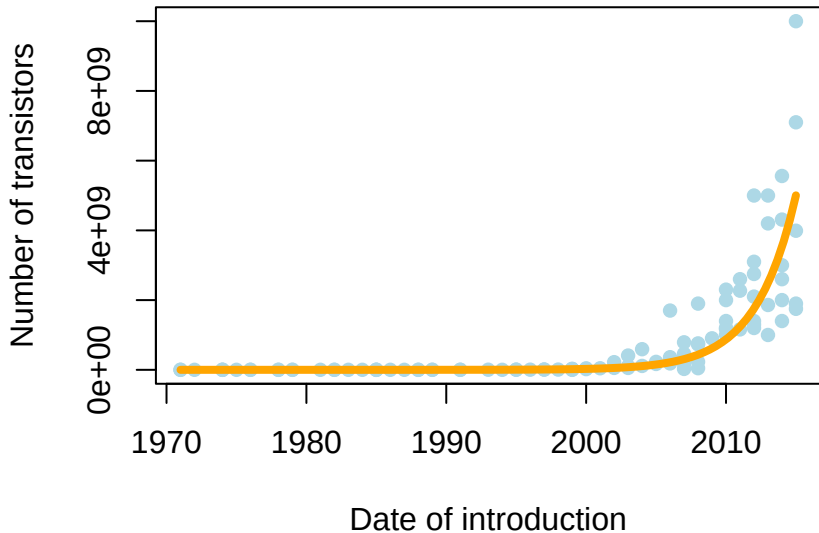
A log-linear model satisfies the linearity assumption

```
log.model <- lm(log(Transistor.count) ~ Date.of.introduction,  
                data=moores.law)  
autoplot(log.model, which=1, ncol=1)
```





Interpreting the log-linear model





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Our model is

$$\log(\widehat{\text{Transistors}}) = -681.21 + 0.35 \cdot \text{Year}$$



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Two interpretations of the slope coefficient:

- Every year, the predicted log of transistors goes up by 0.35



Interpreting the log-linear model

Our model is

$$\log(\widehat{\text{Transistors}}) = -681.21 + 0.35 \cdot \text{Year}$$

Two interpretations of the slope coefficient:

- Every year, the predicted log of transistors goes up by 0.35
- **More useful:** Every year, the predicted number of transistors goes up by 35%

Slope interpretations in the presence of logs

Model	Equation	Interpretation
Linear	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$	1 unit increase in $X \rightarrow \hat{\beta}_1$ unit increase in $\hat{Y}$
Log-linear	$\log(\hat{Y}) = \hat{\beta}_0 + \hat{\beta}_1 X$	1 unit increase in $X \rightarrow \approx (100\hat{\beta}_1)\%$ increase in $\hat{Y}$
Linear-log	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \log(X)$	1% increase in $X \rightarrow \approx 0.01\hat{\beta}_1$ unit increase in $\hat{Y}$
Log-log	$\log(\hat{Y}) = \hat{\beta}_0 + \hat{\beta}_1 \log(X)$	1% increase in $X \rightarrow \approx \hat{\beta}_1\%$ increase in $\hat{Y}$



Slope interpretations in the presence of logs

Model	Example	Interpretation
Linear	$\hat{Y} = 2 + 0.3X$	1 unit increase in $X \rightarrow$ 0.3 unit increase in $\hat{Y}$
Log-linear	$\log(\hat{Y}) = 2 + 0.3X$	1 unit increase in $X \rightarrow$ $\approx 30\%$ increase in $\hat{Y}$
Linear-log	$\hat{Y} = 2 + 0.3 \log(X)$	1% increase in $X \rightarrow$ ≈ 0.003 unit increase in $\hat{Y}$
Log-log	$\log(\hat{Y}) = 2 + 0.3 \log(X)$	1% increase in $X \rightarrow$ $\approx 0.3\%$ increase in $\hat{Y}$



When is the log transformation useful?

- You can transform X , Y , or both
- Anytime you need to "squash" one of the variables (logs make huge numbers not so big!), try transforming it with a log
- In this case, Transistors is skewed right so it is a good candidate for log
- You may need to do a little bit of trial and error to see what works best
- Other transformations are possible!

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An elasticity interpretation for log-log models

- The beer data set contains sales data for cases of beer sold at a supermarket chain
- PRICE12PK is the price of 12-packs, and CASES12PK is the number of 12-packs sold
- There are also variables for 18-packs and 30-packs



What is price elasticity?

- Price elasticity is the ratio of how much demand (quantity sold) will increase when the price increases:

$$\begin{aligned}\text{elasticity} &= \frac{\% \text{ change in demand}}{\% \text{ change in price}} \\ &= \% \text{ increase in demand when price increases by } 1\%\end{aligned}$$



What is price elasticity?

- Price elasticity is the ratio of how much demand (quantity sold) will increase when the price increases:

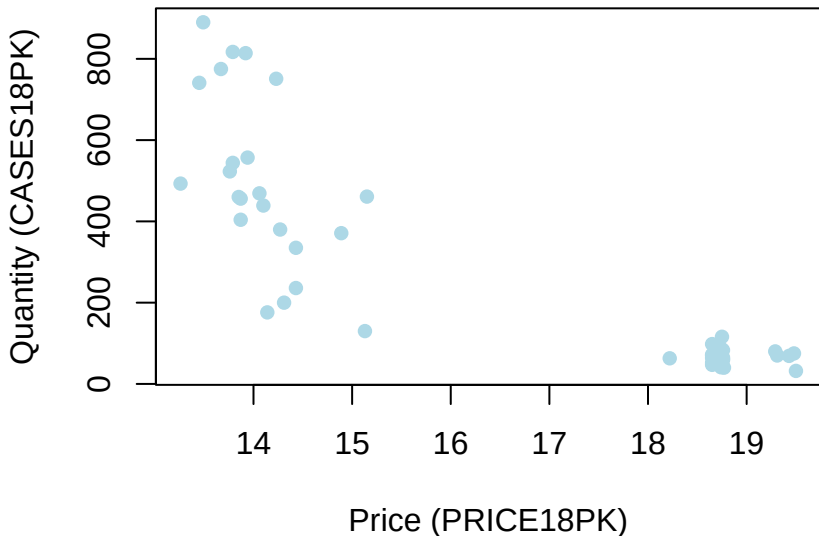
$$\begin{aligned}\text{elasticity} &= \frac{\% \text{ change in demand}}{\% \text{ change in price}} \\ &= \% \text{ increase in demand when price increases by } 1\%\end{aligned}$$

- This is exactly our interpretation of slope coefficients in a log-log model!



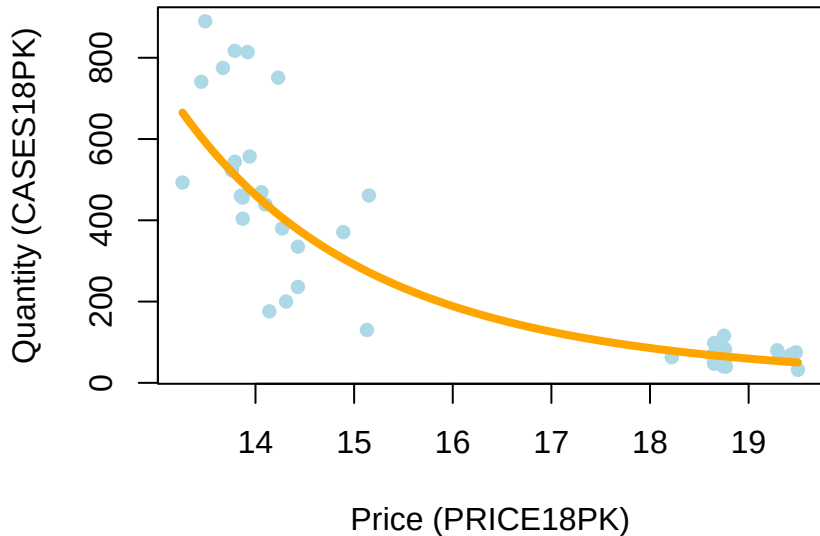
Will a linear model work to predict quantity from price?

This is essentially a demand curve:





Fitting a log-log model





Log transformations can also fix heteroscedasticity!

Compare the residual plots for the linear model (left) vs the log-log model (right):

